

than 2000 tokens per second, are increasingly inadequate for real-time, large-scale AI applications.

Eliminating data movement through architectural convergence

HC1 addresses this challenge with a radical architectural approach: eliminating data movement entirely by removing the separation between compute and memory.

In the HC1, the full Llama 3.1 8B model is hardwired directly into the chip using a mask-ROM-based recall fabric. This design ensures that model weights are permanently embedded within the silicon, removing the need to fetch data from external memory during inference. Key results:

- Sub-millisecond latency
- Dramatically improved energy efficiency
- No dependence on external memory hierarchy

A crucial but often overlooked implication is that this architecture converts memory access into constant-time lookup, eliminating cache hierarchies, DRAM stalls, and bandwidth contention entirely, one of the deepest sources of unpredictability in modern AI systems.

What makes the HC1 unique

The HC1 architecture is built around a unified compute-memory cell, in which a single transistor can store

Taalas—Where AI becomes silicon

Taalas is a US-based semiconductor startup founded in 2023. The company is redefining AI computing through its model-as-silicon architecture, in which neural networks are integrated directly into computer chips rather than run through software. The company's flagship product, the HC1, employs a compute-in-memory design that eliminates data movement, delivers superior performance per watt, and simplifies deployment by using standard air cooling rather than a high-bandwidth memory infrastructure.

By locking down much of the model while retaining only an adaptable layer, Taalas has struck a balance between extreme efficiency and practical deployment. Its strategy centres on two key areas: first, compiling AI models directly into silicon while removing memory bottlenecks to accelerate performance; and second, delivering high-efficiency inference hardware supported by simplified, scalable AI infrastructure.

Taalas represents a broader shift towards purpose-built intelligence hardware, where chips are designed around AI models rather than the other way around.

data and perform multiplication simultaneously. This integration effectively collapses traditionally separate components, such as memory arrays, multiply-accumulate (MAC) units, and interconnect layers, into a single structure.

A key quantitative insight: instead of shuttling weights across millimetres of interconnects, as in GPUs, computation happens in place at the bit-cell level, reducing data movement distance by orders of magnitude, from millimetres to nanometres.

Complementing this is the mask-ROM recall fabric, in which model weights are encoded directly during chip fabrication. This approach enables:

- Multi-bit storage (~4 bits per cell typical)
- Near-zero read energy
- Deterministic latency (no cache

misses, no bandwidth contention)

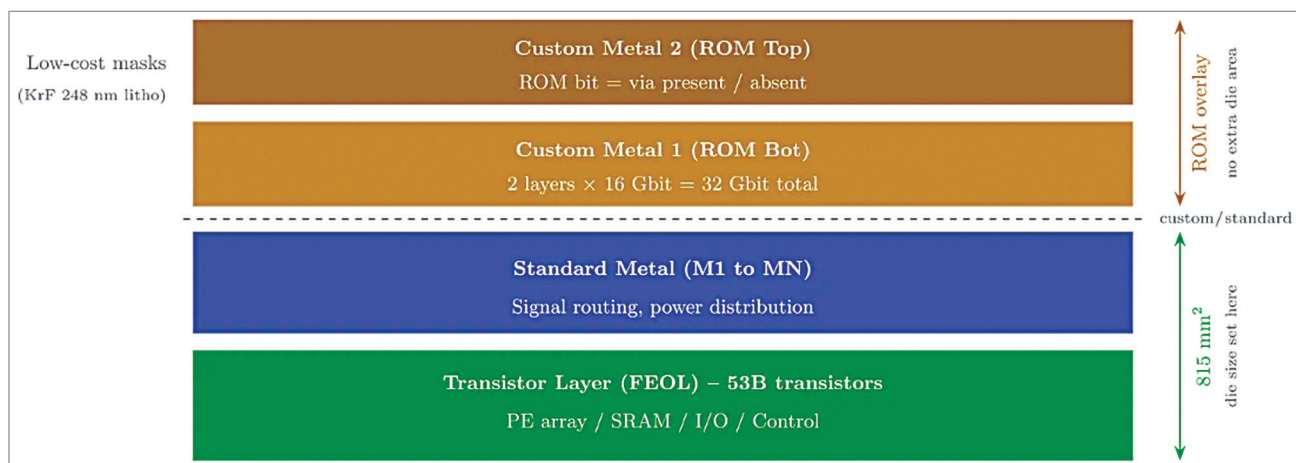
Unlike SRAM or DRAM, mask ROM has effectively zero standby power and extremely high density, making it ideal for inference workloads where weights remain static.

The hybrid layer: Where flexibility still exists

Despite being largely fixed, HC1 incorporates a hybrid dynamic layer, which is critical to its practicality. This SRAM-based subsystem supports:

- KV caching for transformer inference
- LoRA-style fine-tuning overlays
- Adjustable context windows

This is a subtle but crucial design choice. Rather than making the entire model programmable, HC1 keeps 95% of computation fixed and optimises the remaining



HC1 chip vertical cross-section